CA1 – Human Centered Deep Learning

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Section 6: Grid Search

To better optimize our neural network's performance, a comprehensive grid search is conducted using KerasClassifier with GridSearchCV from Scikit-learn. My aim was to systematically evaluate the impact of various batch sizes, optimizers and epoch values while leveraging Henormal weight initialization and early stopping to prevent overfitting.

In [1]:
```
import tensorflow as tf
print("TensorFlow version:", tf.__version__)
```

```
TensorFlow version: 2.12.0
```

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.initializers import HeNormal
from tensorflow.keras.regularizers import l2
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
from tensorflow.keras.wrappers.scikit_learn import KerasClassifier
```

In [3]:
```
# 1. Load and preprocess data
# ----------------------------
# Loading the dataset
df = pd.read_csv("X00231080.csv")

# Split features and target
X = df.drop("Diabetes_binary", axis=1)
y = df["Diabetes_binary"]

# Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train-test split with stratification
seed = 1
np.random.seed(seed)
tf.random.set_seed(seed)

X_train, X_test, y_train, y_test = train_test_split(
```

```
    X_scaled, y, test_size=0.33, random_state=seed, stratify=y
)

print("Training set size:", X_train.shape)
print("Test set size:", X_test.shape)
```

```
Training set size: (28418, 21)
Test set size: (13997, 21)
```

6.1 Experimental Setup

- Model Architecture: 32-16-8 dense layers with ReLU activation, HeNormal initialization, and a sigmoid output layer.
- Dataset Split: 67% training, 33% test with stratified sampling.
- Regularization & Stability:
- EarlyStopping (patience=10, monitored 'loss') to halt training when improvement plateaued.
- ModelCheckpoint to store the best-performing model (based on accuracy).
- Hyperparameter Grid:
  - Batch Sizes: 16, 32, 128, 256
  - Epochs: 30, 50, 100
  - Optimizers: Adam, RMSprop

A total of 24 combinations were evaluated using 3-fold cross-validation on the training set.

In [4]:
```
# 2. Build model function
# ----------------------------
def build_model(optimizer='adam'):
    model = Sequential()
    model.add(Dense(32, activation='relu', input_dim=X_train.shape[1], kernel_initi
    model.add(Dense(16, activation='relu', kernel_initializer=HeNormal()))
    model.add(Dense(8, activation='relu', kernel_initializer=HeNormal()))
    model.add(Dense(1, activation='sigmoid'))

    model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accura
    return model
```

In [5]:
```
# 3. Wrap with KerasClassifier
# ----------------------------
model = KerasClassifier(build_fn=build_model, verbose=0)
```

```
C:\Users\suzak\AppData\Local\Temp\ipykernel_15344\1504816404.py:3: DeprecationWarnin
g: KerasClassifier is deprecated, use Sci-Keras (https://github.com/adriangb/scikera
s) instead. See https://www.adriangb.com/scikeras/stable/migration.html for help mig
rating.
  model = KerasClassifier(build_fn=build_model, verbose=0)
```

In [6]:
```
# 4. Set hyperparameter grid
# ----------------------------
param_grid = {
    'batch_size': [16, 32, 128, 256],
```

```
    'epochs': [30, 50, 100],
    'optimizer': ['adam', 'rmsprop']
}
```

In [7]:
```
# 5. Define callbacks
# ----------------------------
early_stop = EarlyStopping(monitor='loss', patience=10, restore_best_weights=True)
model_checkpoint = ModelCheckpoint("best_model.h5", monitor='accuracy', save_best_o
callbacks_list = [early_stop, model_checkpoint]
```

In [8]:
```
# 6. Run Grid Search
# ----------------------------
grid = GridSearchCV(estimator=model, param_grid=param_grid, cv=3, verbose=2)
grid_result = grid.fit(X_train, y_train, callbacks=callbacks_list)
```

```
Fitting 3 folds for each of 24 candidates, totalling 72 fits
Epoch 1: accuracy improved from -inf to 0.70947, saving model to best_model.h5
Epoch 2: accuracy improved from 0.70947 to 0.74294, saving model to best_model.h5
Epoch 3: accuracy improved from 0.74294 to 0.74706, saving model to best_model.h5
Epoch 4: accuracy improved from 0.74706 to 0.75044, saving model to best_model.h5
Epoch 5: accuracy improved from 0.75044 to 0.75070, saving model to best_model.h5
Epoch 6: accuracy improved from 0.75070 to 0.75260, saving model to best_model.h5
Epoch 7: accuracy improved from 0.75260 to 0.75366, saving model to best_model.h5
Epoch 8: accuracy improved from 0.75366 to 0.75571, saving model to best_model.h5
Epoch 9: accuracy improved from 0.75571 to 0.75629, saving model to best_model.h5
Epoch 10: accuracy improved from 0.75629 to 0.75809, saving model to best_model.h5
Epoch 11: accuracy improved from 0.75809 to 0.75888, saving model to best_model.h5
Epoch 12: accuracy did not improve from 0.75888
Epoch 13: accuracy improved from 0.75888 to 0.75914, saving model to best_model.h5
Epoch 14: accuracy improved from 0.75914 to 0.76141, saving model to best_model.h5
Epoch 15: accuracy improved from 0.76141 to 0.76152, saving model to best_model.h5
Epoch 16: accuracy improved from 0.76152 to 0.76163, saving model to best_model.h5
Epoch 17: accuracy did not improve from 0.76163
Epoch 18: accuracy improved from 0.76163 to 0.76363, saving model to best_model.h5
Epoch 19: accuracy did not improve from 0.76363
Epoch 20: accuracy did not improve from 0.76363
Epoch 21: accuracy improved from 0.76363 to 0.76617, saving model to best_model.h5
Epoch 22: accuracy improved from 0.76617 to 0.76690, saving model to best_model.h5
Epoch 23: accuracy did not improve from 0.76690
Epoch 24: accuracy improved from 0.76690 to 0.76775, saving model to best_model.h5
Epoch 25: accuracy did not improve from 0.76775
Epoch 26: accuracy improved from 0.76775 to 0.76970, saving model to best_model.h5
Epoch 27: accuracy did not improve from 0.76970
```

```
Epoch 28: accuracy did not improve from 0.76970

Epoch 29: accuracy did not improve from 0.76970

Epoch 30: accuracy improved from 0.76970 to 0.77165, saving model to best_model.h5
[CV] END ..........batch_size=16, epochs=30, optimizer=adam; total time=  19.9s

Epoch 1: accuracy did not improve from 0.77165

Epoch 2: accuracy did not improve from 0.77165

Epoch 3: accuracy did not improve from 0.77165

Epoch 4: accuracy did not improve from 0.77165

Epoch 5: accuracy did not improve from 0.77165

Epoch 6: accuracy did not improve from 0.77165

Epoch 7: accuracy did not improve from 0.77165

Epoch 8: accuracy did not improve from 0.77165

Epoch 9: accuracy did not improve from 0.77165

Epoch 10: accuracy did not improve from 0.77165

Epoch 11: accuracy did not improve from 0.77165

Epoch 12: accuracy did not improve from 0.77165

Epoch 13: accuracy did not improve from 0.77165

Epoch 14: accuracy did not improve from 0.77165

Epoch 15: accuracy did not improve from 0.77165

Epoch 16: accuracy did not improve from 0.77165

Epoch 17: accuracy did not improve from 0.77165

Epoch 18: accuracy did not improve from 0.77165

Epoch 19: accuracy did not improve from 0.77165

Epoch 20: accuracy did not improve from 0.77165

Epoch 21: accuracy did not improve from 0.77165

Epoch 22: accuracy did not improve from 0.77165

Epoch 23: accuracy did not improve from 0.77165

Epoch 24: accuracy did not improve from 0.77165

Epoch 25: accuracy did not improve from 0.77165
```

```
Epoch 26: accuracy improved from 0.77165 to 0.77361, saving model to best_model.h5

Epoch 27: accuracy did not improve from 0.77361

Epoch 28: accuracy did not improve from 0.77361

Epoch 29: accuracy improved from 0.77361 to 0.77551, saving model to best_model.h5

Epoch 30: accuracy did not improve from 0.77551
[CV] END ..........batch_size=16, epochs=30, optimizer=adam; total time=  19.2s

Epoch 1: accuracy did not improve from 0.77551

Epoch 2: accuracy did not improve from 0.77551

Epoch 3: accuracy did not improve from 0.77551

Epoch 4: accuracy did not improve from 0.77551

Epoch 5: accuracy did not improve from 0.77551

Epoch 6: accuracy did not improve from 0.77551

Epoch 7: accuracy did not improve from 0.77551

Epoch 8: accuracy did not improve from 0.77551

Epoch 9: accuracy did not improve from 0.77551

Epoch 10: accuracy did not improve from 0.77551

Epoch 11: accuracy did not improve from 0.77551

Epoch 12: accuracy did not improve from 0.77551

Epoch 13: accuracy did not improve from 0.77551

Epoch 14: accuracy did not improve from 0.77551

Epoch 15: accuracy did not improve from 0.77551

Epoch 16: accuracy did not improve from 0.77551

Epoch 17: accuracy did not improve from 0.77551

Epoch 18: accuracy did not improve from 0.77551

Epoch 19: accuracy did not improve from 0.77551

Epoch 20: accuracy did not improve from 0.77551

Epoch 21: accuracy did not improve from 0.77551

Epoch 22: accuracy did not improve from 0.77551
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Epoch 23: accuracy did not improve from 0.77551

Epoch 24: accuracy did not improve from 0.77551

Epoch 25: accuracy did not improve from 0.77551

Epoch 26: accuracy did not improve from 0.77551

Epoch 27: accuracy did not improve from 0.77551

Epoch 28: accuracy did not improve from 0.77551

Epoch 29: accuracy did not improve from 0.77551

Epoch 30: accuracy did not improve from 0.77551
[CV] END ..........batch_size=16, epochs=30, optimizer=adam; total time=  18.9s

Epoch 1: accuracy did not improve from 0.77551

Epoch 2: accuracy did not improve from 0.77551

Epoch 3: accuracy did not improve from 0.77551

Epoch 4: accuracy did not improve from 0.77551

Epoch 5: accuracy did not improve from 0.77551

Epoch 6: accuracy did not improve from 0.77551

Epoch 7: accuracy did not improve from 0.77551

Epoch 8: accuracy did not improve from 0.77551

Epoch 9: accuracy did not improve from 0.77551

Epoch 10: accuracy did not improve from 0.77551

Epoch 11: accuracy did not improve from 0.77551

Epoch 12: accuracy did not improve from 0.77551

Epoch 13: accuracy did not improve from 0.77551

Epoch 14: accuracy did not improve from 0.77551

Epoch 15: accuracy did not improve from 0.77551

Epoch 16: accuracy did not improve from 0.77551

Epoch 17: accuracy did not improve from 0.77551

Epoch 18: accuracy did not improve from 0.77551

Epoch 19: accuracy did not improve from 0.77551

Epoch 20: accuracy did not improve from 0.77551
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Epoch 21: accuracy did not improve from 0.77551

Epoch 22: accuracy did not improve from 0.77551

Epoch 23: accuracy did not improve from 0.77551

Epoch 24: accuracy did not improve from 0.77551

Epoch 25: accuracy did not improve from 0.77551

Epoch 26: accuracy did not improve from 0.77551

Epoch 27: accuracy did not improve from 0.77551

Epoch 28: accuracy did not improve from 0.77551

Epoch 29: accuracy did not improve from 0.77551

Epoch 30: accuracy did not improve from 0.77551
[CV] END ........batch_size=16, epochs=30, optimizer=rmsprop; total time=  18.3s

Epoch 1: accuracy did not improve from 0.77551

Epoch 2: accuracy did not improve from 0.77551

Epoch 3: accuracy did not improve from 0.77551

Epoch 4: accuracy did not improve from 0.77551

Epoch 5: accuracy did not improve from 0.77551

Epoch 6: accuracy did not improve from 0.77551

Epoch 7: accuracy did not improve from 0.77551

Epoch 8: accuracy did not improve from 0.77551

Epoch 9: accuracy did not improve from 0.77551

Epoch 10: accuracy did not improve from 0.77551

Epoch 11: accuracy did not improve from 0.77551

Epoch 12: accuracy did not improve from 0.77551

Epoch 13: accuracy did not improve from 0.77551

Epoch 14: accuracy did not improve from 0.77551

Epoch 15: accuracy did not improve from 0.77551

Epoch 16: accuracy did not improve from 0.77551

Epoch 17: accuracy did not improve from 0.77551
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Epoch 18: accuracy did not improve from 0.77551

Epoch 19: accuracy did not improve from 0.77551

Epoch 20: accuracy did not improve from 0.77551

Epoch 21: accuracy did not improve from 0.77551

Epoch 22: accuracy did not improve from 0.77551

Epoch 23: accuracy did not improve from 0.77551

Epoch 24: accuracy did not improve from 0.77551

Epoch 25: accuracy did not improve from 0.77551

Epoch 26: accuracy did not improve from 0.77551

Epoch 27: accuracy did not improve from 0.77551

Epoch 28: accuracy did not improve from 0.77551

Epoch 29: accuracy did not improve from 0.77551

Epoch 30: accuracy did not improve from 0.77551
[CV] END ........batch_size=16, epochs=30, optimizer=rmsprop; total time=  18.3s

Epoch 1: accuracy did not improve from 0.77551

Epoch 2: accuracy did not improve from 0.77551

Epoch 3: accuracy did not improve from 0.77551

Epoch 4: accuracy did not improve from 0.77551

Epoch 5: accuracy did not improve from 0.77551

Epoch 6: accuracy did not improve from 0.77551

Epoch 7: accuracy did not improve from 0.77551

Epoch 8: accuracy did not improve from 0.77551

Epoch 9: accuracy did not improve from 0.77551

Epoch 10: accuracy did not improve from 0.77551

Epoch 11: accuracy did not improve from 0.77551

Epoch 12: accuracy did not improve from 0.77551

Epoch 13: accuracy did not improve from 0.77551

Epoch 14: accuracy did not improve from 0.77551

Epoch 15: accuracy did not improve from 0.77551
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Epoch 16: accuracy did not improve from 0.77551

Epoch 17: accuracy did not improve from 0.77551

Epoch 18: accuracy did not improve from 0.77551

Epoch 19: accuracy did not improve from 0.77551

Epoch 20: accuracy did not improve from 0.77551

Epoch 21: accuracy did not improve from 0.77551

Epoch 22: accuracy did not improve from 0.77551

Epoch 23: accuracy did not improve from 0.77551

Epoch 24: accuracy did not improve from 0.77551

Epoch 25: accuracy did not improve from 0.77551

Epoch 26: accuracy did not improve from 0.77551

Epoch 27: accuracy did not improve from 0.77551

Epoch 28: accuracy did not improve from 0.77551

Epoch 29: accuracy did not improve from 0.77551

Epoch 30: accuracy did not improve from 0.77551
[CV] END ........batch_size=16, epochs=30, optimizer=rmsprop; total time=  18.3s

Epoch 1: accuracy did not improve from 0.77551

Epoch 2: accuracy did not improve from 0.77551

Epoch 3: accuracy did not improve from 0.77551

Epoch 4: accuracy did not improve from 0.77551

Epoch 5: accuracy did not improve from 0.77551

Epoch 6: accuracy did not improve from 0.77551

Epoch 7: accuracy did not improve from 0.77551

Epoch 8: accuracy did not improve from 0.77551

Epoch 9: accuracy did not improve from 0.77551

Epoch 10: accuracy did not improve from 0.77551

Epoch 11: accuracy did not improve from 0.77551

Epoch 12: accuracy did not improve from 0.77551
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Epoch 13: accuracy did not improve from 0.77551

Epoch 14: accuracy did not improve from 0.77551

Epoch 15: accuracy did not improve from 0.77551

Epoch 16: accuracy did not improve from 0.77551

Epoch 17: accuracy did not improve from 0.77551

Epoch 18: accuracy did not improve from 0.77551

Epoch 19: accuracy did not improve from 0.77551

Epoch 20: accuracy did not improve from 0.77551

Epoch 21: accuracy did not improve from 0.77551

Epoch 22: accuracy did not improve from 0.77551

Epoch 23: accuracy did not improve from 0.77551

Epoch 24: accuracy did not improve from 0.77551

Epoch 25: accuracy did not improve from 0.77551

Epoch 26: accuracy did not improve from 0.77551

Epoch 27: accuracy did not improve from 0.77551

Epoch 28: accuracy did not improve from 0.77551

Epoch 29: accuracy did not improve from 0.77551

Epoch 30: accuracy did not improve from 0.77551

Epoch 31: accuracy did not improve from 0.77551

Epoch 32: accuracy did not improve from 0.77551

Epoch 33: accuracy did not improve from 0.77551

Epoch 34: accuracy did not improve from 0.77551

Epoch 35: accuracy did not improve from 0.77551

Epoch 36: accuracy did not improve from 0.77551

Epoch 37: accuracy did not improve from 0.77551

Epoch 38: accuracy did not improve from 0.77551

Epoch 39: accuracy did not improve from 0.77551

Epoch 40: accuracy did not improve from 0.77551
```

```
Epoch 41: accuracy did not improve from 0.77551

Epoch 42: accuracy did not improve from 0.77551

Epoch 43: accuracy did not improve from 0.77551

Epoch 44: accuracy did not improve from 0.77551

Epoch 45: accuracy did not improve from 0.77551

Epoch 46: accuracy improved from 0.77551 to 0.77577, saving model to best_model.h5

Epoch 47: accuracy improved from 0.77577 to 0.77725, saving model to best_model.h5

Epoch 48: accuracy did not improve from 0.77725

Epoch 49: accuracy did not improve from 0.77725

Epoch 50: accuracy did not improve from 0.77725
[CV] END ..........batch_size=16, epochs=50, optimizer=adam; total time=  31.3s

Epoch 1: accuracy did not improve from 0.77725

Epoch 2: accuracy did not improve from 0.77725

Epoch 3: accuracy did not improve from 0.77725

Epoch 4: accuracy did not improve from 0.77725

Epoch 5: accuracy did not improve from 0.77725

Epoch 6: accuracy did not improve from 0.77725

Epoch 7: accuracy did not improve from 0.77725

Epoch 8: accuracy did not improve from 0.77725

Epoch 9: accuracy did not improve from 0.77725

Epoch 10: accuracy did not improve from 0.77725

Epoch 11: accuracy did not improve from 0.77725

Epoch 12: accuracy did not improve from 0.77725

Epoch 13: accuracy did not improve from 0.77725

Epoch 14: accuracy did not improve from 0.77725

Epoch 15: accuracy did not improve from 0.77725

Epoch 16: accuracy did not improve from 0.77725

Epoch 17: accuracy did not improve from 0.77725

Epoch 18: accuracy did not improve from 0.77725
```

```
Epoch 19: accuracy did not improve from 0.77725

Epoch 20: accuracy did not improve from 0.77725

Epoch 21: accuracy did not improve from 0.77725

Epoch 22: accuracy did not improve from 0.77725

Epoch 23: accuracy did not improve from 0.77725

Epoch 24: accuracy did not improve from 0.77725

Epoch 25: accuracy did not improve from 0.77725

Epoch 26: accuracy did not improve from 0.77725

Epoch 27: accuracy did not improve from 0.77725

Epoch 28: accuracy improved from 0.77725 to 0.77730, saving model to best_model.h5

Epoch 29: accuracy did not improve from 0.77730

Epoch 30: accuracy improved from 0.77730 to 0.77767, saving model to best_model.h5

Epoch 31: accuracy improved from 0.77767 to 0.77778, saving model to best_model.h5

Epoch 32: accuracy improved from 0.77778 to 0.77815, saving model to best_model.h5

Epoch 33: accuracy did not improve from 0.77815

Epoch 34: accuracy improved from 0.77815 to 0.77825, saving model to best_model.h5

Epoch 35: accuracy improved from 0.77825 to 0.77941, saving model to best_model.h5

Epoch 36: accuracy did not improve from 0.77941

Epoch 37: accuracy improved from 0.77941 to 0.78036, saving model to best_model.h5

Epoch 38: accuracy did not improve from 0.78036

Epoch 39: accuracy did not improve from 0.78036

Epoch 40: accuracy did not improve from 0.78036

Epoch 41: accuracy improved from 0.78036 to 0.78158, saving model to best_model.h5

Epoch 42: accuracy improved from 0.78158 to 0.78358, saving model to best_model.h5

Epoch 43: accuracy did not improve from 0.78358

Epoch 44: accuracy did not improve from 0.78358

Epoch 45: accuracy did not improve from 0.78358

Epoch 46: accuracy did not improve from 0.78358
```

```
Epoch 47: accuracy did not improve from 0.78358

Epoch 48: accuracy did not improve from 0.78358

Epoch 49: accuracy did not improve from 0.78358

Epoch 50: accuracy did not improve from 0.78358
[CV] END ...........batch_size=16, epochs=50, optimizer=adam; total time=  31.5s

Epoch 1: accuracy did not improve from 0.78358

Epoch 2: accuracy did not improve from 0.78358

Epoch 3: accuracy did not improve from 0.78358

Epoch 4: accuracy did not improve from 0.78358

Epoch 5: accuracy did not improve from 0.78358

Epoch 6: accuracy did not improve from 0.78358

Epoch 7: accuracy did not improve from 0.78358

Epoch 8: accuracy did not improve from 0.78358

Epoch 9: accuracy did not improve from 0.78358

Epoch 10: accuracy did not improve from 0.78358

Epoch 11: accuracy did not improve from 0.78358

Epoch 12: accuracy did not improve from 0.78358

Epoch 13: accuracy did not improve from 0.78358

Epoch 14: accuracy did not improve from 0.78358

Epoch 15: accuracy did not improve from 0.78358

Epoch 16: accuracy did not improve from 0.78358

Epoch 17: accuracy did not improve from 0.78358

Epoch 18: accuracy did not improve from 0.78358

Epoch 19: accuracy did not improve from 0.78358

Epoch 20: accuracy did not improve from 0.78358

Epoch 21: accuracy did not improve from 0.78358

Epoch 22: accuracy did not improve from 0.78358

Epoch 23: accuracy did not improve from 0.78358
```

```
Epoch 24: accuracy did not improve from 0.78358

Epoch 25: accuracy did not improve from 0.78358

Epoch 26: accuracy did not improve from 0.78358

Epoch 27: accuracy did not improve from 0.78358

Epoch 28: accuracy did not improve from 0.78358

Epoch 29: accuracy did not improve from 0.78358

Epoch 30: accuracy did not improve from 0.78358

Epoch 31: accuracy did not improve from 0.78358

Epoch 32: accuracy did not improve from 0.78358

Epoch 33: accuracy did not improve from 0.78358

Epoch 34: accuracy did not improve from 0.78358

Epoch 35: accuracy did not improve from 0.78358

Epoch 36: accuracy did not improve from 0.78358

Epoch 37: accuracy did not improve from 0.78358

Epoch 38: accuracy did not improve from 0.78358

Epoch 39: accuracy did not improve from 0.78358

Epoch 40: accuracy did not improve from 0.78358

Epoch 41: accuracy did not improve from 0.78358

Epoch 42: accuracy did not improve from 0.78358

Epoch 43: accuracy did not improve from 0.78358

Epoch 44: accuracy did not improve from 0.78358

Epoch 45: accuracy did not improve from 0.78358

Epoch 46: accuracy did not improve from 0.78358

Epoch 47: accuracy did not improve from 0.78358

Epoch 48: accuracy did not improve from 0.78358

Epoch 49: accuracy did not improve from 0.78358

Epoch 50: accuracy did not improve from 0.78358
[CV] END ..........batch_size=16, epochs=50, optimizer=adam; total time=  31.2s

Epoch 1: accuracy did not improve from 0.78358
```

```
Epoch 2: accuracy did not improve from 0.78358

Epoch 3: accuracy did not improve from 0.78358

Epoch 4: accuracy did not improve from 0.78358

Epoch 5: accuracy did not improve from 0.78358

Epoch 6: accuracy did not improve from 0.78358

Epoch 7: accuracy did not improve from 0.78358

Epoch 8: accuracy did not improve from 0.78358

Epoch 9: accuracy did not improve from 0.78358

Epoch 10: accuracy did not improve from 0.78358

Epoch 11: accuracy did not improve from 0.78358

Epoch 12: accuracy did not improve from 0.78358

Epoch 13: accuracy did not improve from 0.78358

Epoch 14: accuracy did not improve from 0.78358

Epoch 15: accuracy did not improve from 0.78358

Epoch 16: accuracy did not improve from 0.78358

Epoch 17: accuracy did not improve from 0.78358

Epoch 18: accuracy did not improve from 0.78358

Epoch 19: accuracy did not improve from 0.78358

Epoch 20: accuracy did not improve from 0.78358

Epoch 21: accuracy did not improve from 0.78358

Epoch 22: accuracy did not improve from 0.78358

Epoch 23: accuracy did not improve from 0.78358

Epoch 24: accuracy did not improve from 0.78358

Epoch 25: accuracy did not improve from 0.78358

Epoch 26: accuracy did not improve from 0.78358

Epoch 27: accuracy did not improve from 0.78358

Epoch 28: accuracy did not improve from 0.78358

Epoch 29: accuracy did not improve from 0.78358
```

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Epoch 30: accuracy did not improve from 0.78358

Epoch 31: accuracy did not improve from 0.78358

Epoch 32: accuracy did not improve from 0.78358

Epoch 33: accuracy did not improve from 0.78358

Epoch 34: accuracy did not improve from 0.78358

Epoch 35: accuracy did not improve from 0.78358

Epoch 36: accuracy did not improve from 0.78358

Epoch 37: accuracy did not improve from 0.78358

Epoch 38: accuracy did not improve from 0.78358

Epoch 39: accuracy did not improve from 0.78358

Epoch 40: accuracy did not improve from 0.78358

Epoch 41: accuracy did not improve from 0.78358

Epoch 42: accuracy did not improve from 0.78358

Epoch 43: accuracy did not improve from 0.78358

Epoch 44: accuracy did not improve from 0.78358

Epoch 45: accuracy did not improve from 0.78358

Epoch 46: accuracy did not improve from 0.78358

Epoch 47: accuracy did not improve from 0.78358

Epoch 48: accuracy did not improve from 0.78358

Epoch 49: accuracy did not improve from 0.78358

Epoch 50: accuracy did not improve from 0.78358
[CV] END ........batch_size=16, epochs=50, optimizer=rmsprop; total time=  30.2s

Epoch 1: accuracy did not improve from 0.78358

Epoch 2: accuracy did not improve from 0.78358

Epoch 3: accuracy did not improve from 0.78358

Epoch 4: accuracy did not improve from 0.78358

Epoch 5: accuracy did not improve from 0.78358

Epoch 6: accuracy did not improve from 0.78358
```

```
Epoch 7: accuracy did not improve from 0.78358

Epoch 8: accuracy did not improve from 0.78358

Epoch 9: accuracy did not improve from 0.78358

Epoch 10: accuracy did not improve from 0.78358

Epoch 11: accuracy did not improve from 0.78358

Epoch 12: accuracy did not improve from 0.78358

Epoch 13: accuracy did not improve from 0.78358

Epoch 14: accuracy did not improve from 0.78358

Epoch 15: accuracy did not improve from 0.78358

Epoch 16: accuracy did not improve from 0.78358

Epoch 17: accuracy did not improve from 0.78358

Epoch 18: accuracy did not improve from 0.78358

Epoch 19: accuracy did not improve from 0.78358

Epoch 20: accuracy did not improve from 0.78358

Epoch 21: accuracy did not improve from 0.78358

Epoch 22: accuracy did not improve from 0.78358

Epoch 23: accuracy did not improve from 0.78358

Epoch 24: accuracy did not improve from 0.78358

Epoch 25: accuracy did not improve from 0.78358

Epoch 26: accuracy did not improve from 0.78358

Epoch 27: accuracy did not improve from 0.78358

Epoch 28: accuracy did not improve from 0.78358

Epoch 29: accuracy did not improve from 0.78358

Epoch 30: accuracy did not improve from 0.78358

Epoch 31: accuracy did not improve from 0.78358

Epoch 32: accuracy did not improve from 0.78358

Epoch 33: accuracy did not improve from 0.78358

Epoch 34: accuracy did not improve from 0.78358
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Epoch 35: accuracy did not improve from 0.78358

Epoch 36: accuracy did not improve from 0.78358

Epoch 37: accuracy did not improve from 0.78358

Epoch 38: accuracy did not improve from 0.78358
[CV] END ........batch_size=16, epochs=50, optimizer=rmsprop; total time=  23.8s

Epoch 1: accuracy did not improve from 0.78358

Epoch 2: accuracy did not improve from 0.78358

Epoch 3: accuracy did not improve from 0.78358

Epoch 4: accuracy did not improve from 0.78358

Epoch 5: accuracy did not improve from 0.78358

Epoch 6: accuracy did not improve from 0.78358

Epoch 7: accuracy did not improve from 0.78358

Epoch 8: accuracy did not improve from 0.78358

Epoch 9: accuracy did not improve from 0.78358

Epoch 10: accuracy did not improve from 0.78358

Epoch 11: accuracy did not improve from 0.78358

Epoch 12: accuracy did not improve from 0.78358

Epoch 13: accuracy did not improve from 0.78358

Epoch 14: accuracy did not improve from 0.78358

Epoch 15: accuracy did not improve from 0.78358

Epoch 16: accuracy did not improve from 0.78358

Epoch 17: accuracy did not improve from 0.78358

Epoch 18: accuracy did not improve from 0.78358

Epoch 19: accuracy did not improve from 0.78358

Epoch 20: accuracy did not improve from 0.78358

Epoch 21: accuracy did not improve from 0.78358

Epoch 22: accuracy did not improve from 0.78358

Epoch 23: accuracy did not improve from 0.78358

Epoch 24: accuracy did not improve from 0.78358
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Epoch 25: accuracy did not improve from 0.78358

Epoch 26: accuracy did not improve from 0.78358

Epoch 27: accuracy did not improve from 0.78358

Epoch 28: accuracy did not improve from 0.78358

Epoch 29: accuracy did not improve from 0.78358

Epoch 30: accuracy did not improve from 0.78358

Epoch 31: accuracy did not improve from 0.78358

Epoch 32: accuracy did not improve from 0.78358

Epoch 33: accuracy did not improve from 0.78358

Epoch 34: accuracy did not improve from 0.78358

Epoch 35: accuracy did not improve from 0.78358

Epoch 36: accuracy did not improve from 0.78358

Epoch 37: accuracy did not improve from 0.78358

Epoch 38: accuracy did not improve from 0.78358

Epoch 39: accuracy did not improve from 0.78358

Epoch 40: accuracy did not improve from 0.78358

Epoch 41: accuracy did not improve from 0.78358

Epoch 42: accuracy did not improve from 0.78358

Epoch 43: accuracy did not improve from 0.78358

Epoch 44: accuracy did not improve from 0.78358

Epoch 45: accuracy did not improve from 0.78358

Epoch 46: accuracy did not improve from 0.78358

Epoch 47: accuracy did not improve from 0.78358

Epoch 48: accuracy did not improve from 0.78358

Epoch 49: accuracy did not improve from 0.78358

Epoch 50: accuracy did not improve from 0.78358
[CV] END ........batch_size=16, epochs=50, optimizer=rmsprop; total time=  30.3s

Epoch 1: accuracy did not improve from 0.78358
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Epoch 2: accuracy did not improve from 0.78358

Epoch 3: accuracy did not improve from 0.78358

Epoch 4: accuracy did not improve from 0.78358

Epoch 5: accuracy did not improve from 0.78358

Epoch 6: accuracy did not improve from 0.78358

Epoch 7: accuracy did not improve from 0.78358

Epoch 8: accuracy did not improve from 0.78358

Epoch 9: accuracy did not improve from 0.78358

Epoch 10: accuracy did not improve from 0.78358

Epoch 11: accuracy did not improve from 0.78358

Epoch 12: accuracy did not improve from 0.78358

Epoch 13: accuracy did not improve from 0.78358

Epoch 14: accuracy did not improve from 0.78358

Epoch 15: accuracy did not improve from 0.78358

Epoch 16: accuracy did not improve from 0.78358

Epoch 17: accuracy did not improve from 0.78358

Epoch 18: accuracy did not improve from 0.78358

Epoch 19: accuracy did not improve from 0.78358

Epoch 20: accuracy did not improve from 0.78358

Epoch 21: accuracy did not improve from 0.78358

Epoch 22: accuracy did not improve from 0.78358

Epoch 23: accuracy did not improve from 0.78358

Epoch 24: accuracy did not improve from 0.78358

Epoch 25: accuracy did not improve from 0.78358

Epoch 26: accuracy did not improve from 0.78358

Epoch 27: accuracy did not improve from 0.78358

Epoch 28: accuracy did not improve from 0.78358

Epoch 29: accuracy did not improve from 0.78358
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Epoch 30: accuracy did not improve from 0.78358

Epoch 31: accuracy did not improve from 0.78358

Epoch 32: accuracy did not improve from 0.78358

Epoch 33: accuracy did not improve from 0.78358

Epoch 34: accuracy did not improve from 0.78358

Epoch 35: accuracy did not improve from 0.78358

Epoch 36: accuracy did not improve from 0.78358

Epoch 37: accuracy did not improve from 0.78358

Epoch 38: accuracy did not improve from 0.78358

Epoch 39: accuracy did not improve from 0.78358

Epoch 40: accuracy did not improve from 0.78358

Epoch 41: accuracy did not improve from 0.78358

Epoch 42: accuracy did not improve from 0.78358

Epoch 43: accuracy did not improve from 0.78358

Epoch 44: accuracy did not improve from 0.78358

Epoch 45: accuracy did not improve from 0.78358

Epoch 46: accuracy did not improve from 0.78358

Epoch 47: accuracy did not improve from 0.78358

Epoch 48: accuracy did not improve from 0.78358

Epoch 49: accuracy did not improve from 0.78358

Epoch 50: accuracy did not improve from 0.78358

Epoch 51: accuracy did not improve from 0.78358

Epoch 52: accuracy did not improve from 0.78358

Epoch 53: accuracy did not improve from 0.78358

Epoch 54: accuracy did not improve from 0.78358

Epoch 55: accuracy did not improve from 0.78358

Epoch 56: accuracy did not improve from 0.78358

Epoch 57: accuracy did not improve from 0.78358
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Epoch 58: accuracy did not improve from 0.78358

Epoch 59: accuracy did not improve from 0.78358

Epoch 60: accuracy did not improve from 0.78358

Epoch 61: accuracy did not improve from 0.78358

Epoch 62: accuracy did not improve from 0.78358

Epoch 63: accuracy did not improve from 0.78358

Epoch 64: accuracy did not improve from 0.78358

Epoch 65: accuracy did not improve from 0.78358

Epoch 66: accuracy did not improve from 0.78358

Epoch 67: accuracy did not improve from 0.78358

Epoch 68: accuracy did not improve from 0.78358

Epoch 69: accuracy did not improve from 0.78358

Epoch 70: accuracy improved from 0.78358 to 0.78506, saving model to best_model.h5

Epoch 71: accuracy did not improve from 0.78506

Epoch 72: accuracy did not improve from 0.78506

Epoch 73: accuracy did not improve from 0.78506

Epoch 74: accuracy did not improve from 0.78506

Epoch 75: accuracy did not improve from 0.78506

Epoch 76: accuracy improved from 0.78506 to 0.78564, saving model to best_model.h5

Epoch 77: accuracy did not improve from 0.78564

Epoch 78: accuracy did not improve from 0.78564

Epoch 79: accuracy improved from 0.78564 to 0.78617, saving model to best_model.h5

Epoch 80: accuracy did not improve from 0.78617

Epoch 81: accuracy improved from 0.78617 to 0.78760, saving model to best_model.h5

Epoch 82: accuracy did not improve from 0.78760

Epoch 83: accuracy did not improve from 0.78760

Epoch 84: accuracy did not improve from 0.78760

Epoch 85: accuracy did not improve from 0.78760
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Epoch 86: accuracy did not improve from 0.78760

Epoch 87: accuracy did not improve from 0.78760

Epoch 88: accuracy did not improve from 0.78760

Epoch 89: accuracy did not improve from 0.78760

Epoch 90: accuracy did not improve from 0.78760

Epoch 91: accuracy did not improve from 0.78760

Epoch 92: accuracy improved from 0.78760 to 0.79002, saving model to best_model.h5

Epoch 93: accuracy did not improve from 0.79002

Epoch 94: accuracy did not improve from 0.79002

Epoch 95: accuracy did not improve from 0.79002

Epoch 96: accuracy did not improve from 0.79002

Epoch 97: accuracy did not improve from 0.79002

Epoch 98: accuracy did not improve from 0.79002

Epoch 99: accuracy did not improve from 0.79002

Epoch 100: accuracy did not improve from 0.79002
[CV] END ..........batch_size=16, epochs=100, optimizer=adam; total time= 1.0min

Epoch 1: accuracy did not improve from 0.79002

Epoch 2: accuracy did not improve from 0.79002

Epoch 3: accuracy did not improve from 0.79002

Epoch 4: accuracy did not improve from 0.79002

Epoch 5: accuracy did not improve from 0.79002

Epoch 6: accuracy did not improve from 0.79002

Epoch 7: accuracy did not improve from 0.79002

Epoch 8: accuracy did not improve from 0.79002

Epoch 9: accuracy did not improve from 0.79002

Epoch 10: accuracy did not improve from 0.79002

Epoch 11: accuracy did not improve from 0.79002

Epoch 12: accuracy did not improve from 0.79002

Epoch 13: accuracy did not improve from 0.79002
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Epoch 14: accuracy did not improve from 0.79002

Epoch 15: accuracy did not improve from 0.79002

Epoch 16: accuracy did not improve from 0.79002

Epoch 17: accuracy did not improve from 0.79002

Epoch 18: accuracy did not improve from 0.79002

Epoch 19: accuracy did not improve from 0.79002

Epoch 20: accuracy did not improve from 0.79002

Epoch 21: accuracy did not improve from 0.79002

Epoch 22: accuracy did not improve from 0.79002

Epoch 23: accuracy did not improve from 0.79002

Epoch 24: accuracy did not improve from 0.79002

Epoch 25: accuracy did not improve from 0.79002

Epoch 26: accuracy did not improve from 0.79002

Epoch 27: accuracy did not improve from 0.79002

Epoch 28: accuracy did not improve from 0.79002

Epoch 29: accuracy did not improve from 0.79002

Epoch 30: accuracy did not improve from 0.79002

Epoch 31: accuracy did not improve from 0.79002

Epoch 32: accuracy did not improve from 0.79002

Epoch 33: accuracy did not improve from 0.79002

Epoch 34: accuracy did not improve from 0.79002

Epoch 35: accuracy did not improve from 0.79002

Epoch 36: accuracy did not improve from 0.79002

Epoch 37: accuracy did not improve from 0.79002

Epoch 38: accuracy did not improve from 0.79002

Epoch 39: accuracy did not improve from 0.79002

Epoch 40: accuracy did not improve from 0.79002

Epoch 41: accuracy did not improve from 0.79002
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Epoch 42: accuracy did not improve from 0.79002

Epoch 43: accuracy did not improve from 0.79002

Epoch 44: accuracy did not improve from 0.79002

Epoch 45: accuracy did not improve from 0.79002

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Epoch 47: accuracy did not improve from 0.79002

Epoch 48: accuracy did not improve from 0.79002

Epoch 49: accuracy did not improve from 0.79002

Epoch 50: accuracy did not improve from 0.79002

Epoch 51: accuracy did not improve from 0.79002

Epoch 52: accuracy did not improve from 0.79002

Epoch 53: accuracy did not improve from 0.79002

Epoch 54: accuracy did not improve from 0.79002

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Epoch 58: accuracy did not improve from 0.79002

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Epoch 62: accuracy did not improve from 0.79002

Epoch 63: accuracy did not improve from 0.79002

Epoch 64: accuracy did not improve from 0.79002

Epoch 65: accuracy did not improve from 0.79002

Epoch 66: accuracy did not improve from 0.79002

Epoch 67: accuracy did not improve from 0.79002

Epoch 68: accuracy did not improve from 0.79002

Epoch 69: accuracy did not improve from 0.79002
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Epoch 70: accuracy did not improve from 0.79002

Epoch 71: accuracy did not improve from 0.79002

Epoch 72: accuracy did not improve from 0.79002

Epoch 73: accuracy did not improve from 0.79002

Epoch 74: accuracy did not improve from 0.79002

Epoch 75: accuracy did not improve from 0.79002

Epoch 76: accuracy did not improve from 0.79002

Epoch 77: accuracy did not improve from 0.79002

Epoch 78: accuracy did not improve from 0.79002

Epoch 79: accuracy did not improve from 0.79002

Epoch 80: accuracy did not improve from 0.79002

Epoch 81: accuracy did not improve from 0.79002

Epoch 82: accuracy did not improve from 0.79002

Epoch 83: accuracy did not improve from 0.79002

Epoch 84: accuracy did not improve from 0.79002

Epoch 85: accuracy did not improve from 0.79002

Epoch 86: accuracy improved from 0.79002 to 0.79082, saving model to best_model.h5

Epoch 87: accuracy did not improve from 0.79082

Epoch 88: accuracy improved from 0.79082 to 0.79171, saving model to best_model.h5

Epoch 89: accuracy did not improve from 0.79171

Epoch 90: accuracy did not improve from 0.79171

Epoch 91: accuracy did not improve from 0.79171

Epoch 92: accuracy improved from 0.79171 to 0.79356, saving model to best_model.h5

Epoch 93: accuracy did not improve from 0.79356

Epoch 94: accuracy did not improve from 0.79356

Epoch 95: accuracy did not improve from 0.79356

Epoch 96: accuracy did not improve from 0.79356

Epoch 97: accuracy improved from 0.79356 to 0.79367, saving model to best_model.h5
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Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ..........batch_size=16, epochs=100, optimizer=adam; total time= 1.0min

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367
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Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

Epoch 34: accuracy did not improve from 0.79367

Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367

Epoch 51: accuracy did not improve from 0.79367

Epoch 52: accuracy did not improve from 0.79367
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Epoch 53: accuracy did not improve from 0.79367

Epoch 54: accuracy did not improve from 0.79367

Epoch 55: accuracy did not improve from 0.79367

Epoch 56: accuracy did not improve from 0.79367

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Epoch 58: accuracy did not improve from 0.79367

Epoch 59: accuracy did not improve from 0.79367

Epoch 60: accuracy did not improve from 0.79367

Epoch 61: accuracy did not improve from 0.79367

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Epoch 64: accuracy did not improve from 0.79367

Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

Epoch 67: accuracy did not improve from 0.79367

Epoch 68: accuracy did not improve from 0.79367

Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

Epoch 71: accuracy did not improve from 0.79367

Epoch 72: accuracy did not improve from 0.79367

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Epoch 79: accuracy did not improve from 0.79367

Epoch 80: accuracy did not improve from 0.79367
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Epoch 81: accuracy did not improve from 0.79367

Epoch 82: accuracy did not improve from 0.79367

Epoch 83: accuracy did not improve from 0.79367

Epoch 84: accuracy did not improve from 0.79367

Epoch 85: accuracy did not improve from 0.79367

Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

Epoch 88: accuracy did not improve from 0.79367

Epoch 89: accuracy did not improve from 0.79367

Epoch 90: accuracy did not improve from 0.79367

Epoch 91: accuracy did not improve from 0.79367

Epoch 92: accuracy did not improve from 0.79367

Epoch 93: accuracy did not improve from 0.79367

Epoch 94: accuracy did not improve from 0.79367

Epoch 95: accuracy did not improve from 0.79367

Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ..........batch_size=16, epochs=100, optimizer=adam; total time= 1.0min

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367
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Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

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Epoch 27: accuracy did not improve from 0.79367

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Epoch 34: accuracy did not improve from 0.79367

Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367
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Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

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Epoch 63: accuracy did not improve from 0.79367

Epoch 64: accuracy did not improve from 0.79367
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Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

Epoch 67: accuracy did not improve from 0.79367

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Epoch 85: accuracy did not improve from 0.79367

Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

Epoch 88: accuracy did not improve from 0.79367

Epoch 89: accuracy did not improve from 0.79367

Epoch 90: accuracy did not improve from 0.79367

Epoch 91: accuracy did not improve from 0.79367

Epoch 92: accuracy did not improve from 0.79367
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Epoch 93: accuracy did not improve from 0.79367

Epoch 94: accuracy did not improve from 0.79367

Epoch 95: accuracy did not improve from 0.79367

Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END .......batch_size=16, epochs=100, optimizer=rmsprop; total time=  59.4s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

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Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

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Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367
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Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

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Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

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Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367
[CV] END .......batch_size=16, epochs=100, optimizer=rmsprop; total time=  24.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367
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Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

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Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

Epoch 34: accuracy did not improve from 0.79367
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Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

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Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367

Epoch 51: accuracy did not improve from 0.79367

Epoch 52: accuracy did not improve from 0.79367

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Epoch 58: accuracy did not improve from 0.79367

Epoch 59: accuracy did not improve from 0.79367

Epoch 60: accuracy did not improve from 0.79367

Epoch 61: accuracy did not improve from 0.79367

Epoch 62: accuracy did not improve from 0.79367
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Epoch 63: accuracy did not improve from 0.79367

Epoch 64: accuracy did not improve from 0.79367

Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

Epoch 67: accuracy did not improve from 0.79367

Epoch 68: accuracy did not improve from 0.79367

Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

Epoch 71: accuracy did not improve from 0.79367

Epoch 72: accuracy did not improve from 0.79367

Epoch 73: accuracy did not improve from 0.79367

Epoch 74: accuracy did not improve from 0.79367

Epoch 75: accuracy did not improve from 0.79367

Epoch 76: accuracy did not improve from 0.79367

Epoch 77: accuracy did not improve from 0.79367

Epoch 78: accuracy did not improve from 0.79367

Epoch 79: accuracy did not improve from 0.79367

Epoch 80: accuracy did not improve from 0.79367
[CV] END .......batch_size=16, epochs=100, optimizer=rmsprop; total time=  48.0s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367
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Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=30, optimizer=adam; total time=   9.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367
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Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=30, optimizer=adam; total time=   9.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367
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Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=30, optimizer=adam; total time=  10.0s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367
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Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

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Epoch 26: accuracy did not improve from 0.79367

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Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
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[CV] END ........batch_size=32, epochs=30, optimizer=rmsprop; total time=   9.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

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Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367
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Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ........batch_size=32, epochs=30, optimizer=rmsprop; total time=   9.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

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Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367
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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ........batch_size=32, epochs=30, optimizer=rmsprop; total time=   9.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367
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Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

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Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

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Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=50, optimizer=adam; total time=  16.2s
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Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

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Epoch 19: accuracy did not improve from 0.79367

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Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367
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Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

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Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=50, optimizer=adam; total time=  16.0s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367
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Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

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Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367
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Epoch 34: accuracy did not improve from 0.79367

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Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=50, optimizer=adam; total time=  16.2s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367
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Epoch 12: accuracy did not improve from 0.79367

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Epoch 39: accuracy did not improve from 0.79367
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Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

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Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ........batch_size=32, epochs=50, optimizer=rmsprop; total time=  15.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

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Epoch 44: accuracy did not improve from 0.79367
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Epoch 45: accuracy did not improve from 0.79367

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Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ........batch_size=32, epochs=50, optimizer=rmsprop; total time=  15.5s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367
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Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

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Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
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[CV] END ........batch_size=32, epochs=50, optimizer=rmsprop; total time=  15.5s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367
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Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

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Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

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Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367

Epoch 51: accuracy did not improve from 0.79367

Epoch 52: accuracy did not improve from 0.79367

Epoch 53: accuracy did not improve from 0.79367

Epoch 54: accuracy did not improve from 0.79367

Epoch 55: accuracy did not improve from 0.79367
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Epoch 56: accuracy did not improve from 0.79367

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Epoch 58: accuracy did not improve from 0.79367

Epoch 59: accuracy did not improve from 0.79367

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Epoch 62: accuracy did not improve from 0.79367

Epoch 63: accuracy did not improve from 0.79367

Epoch 64: accuracy did not improve from 0.79367

Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

Epoch 67: accuracy did not improve from 0.79367

Epoch 68: accuracy did not improve from 0.79367

Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

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Epoch 80: accuracy did not improve from 0.79367

Epoch 81: accuracy did not improve from 0.79367

Epoch 82: accuracy did not improve from 0.79367

Epoch 83: accuracy did not improve from 0.79367
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Epoch 84: accuracy did not improve from 0.79367

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Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

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Epoch 92: accuracy did not improve from 0.79367

Epoch 93: accuracy did not improve from 0.79367

Epoch 94: accuracy did not improve from 0.79367

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Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=100, optimizer=adam; total time=  31.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367
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Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

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Epoch 31: accuracy did not improve from 0.79367

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Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367
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Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

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Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

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Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367

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Epoch 63: accuracy did not improve from 0.79367

Epoch 64: accuracy did not improve from 0.79367

Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

Epoch 67: accuracy did not improve from 0.79367
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Epoch 68: accuracy did not improve from 0.79367

Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

Epoch 71: accuracy did not improve from 0.79367

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Epoch 86: accuracy did not improve from 0.79367

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Epoch 90: accuracy did not improve from 0.79367

Epoch 91: accuracy did not improve from 0.79367

Epoch 92: accuracy did not improve from 0.79367

Epoch 93: accuracy did not improve from 0.79367

Epoch 94: accuracy did not improve from 0.79367

Epoch 95: accuracy did not improve from 0.79367
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Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=100, optimizer=adam; total time=  32.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367
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Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

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Epoch 30: accuracy did not improve from 0.79367

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Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
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Epoch 51: accuracy did not improve from 0.79367

Epoch 52: accuracy did not improve from 0.79367

Epoch 53: accuracy did not improve from 0.79367

Epoch 54: accuracy did not improve from 0.79367

Epoch 55: accuracy did not improve from 0.79367

Epoch 56: accuracy did not improve from 0.79367

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Epoch 58: accuracy did not improve from 0.79367

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Epoch 61: accuracy did not improve from 0.79367

Epoch 62: accuracy did not improve from 0.79367

Epoch 63: accuracy did not improve from 0.79367

Epoch 64: accuracy did not improve from 0.79367

Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

Epoch 67: accuracy did not improve from 0.79367

Epoch 68: accuracy did not improve from 0.79367

Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

Epoch 71: accuracy did not improve from 0.79367

Epoch 72: accuracy did not improve from 0.79367

Epoch 73: accuracy did not improve from 0.79367

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Epoch 75: accuracy did not improve from 0.79367

Epoch 76: accuracy did not improve from 0.79367

Epoch 77: accuracy did not improve from 0.79367

Epoch 78: accuracy did not improve from 0.79367
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Epoch 79: accuracy did not improve from 0.79367

Epoch 80: accuracy did not improve from 0.79367

Epoch 81: accuracy did not improve from 0.79367

Epoch 82: accuracy did not improve from 0.79367

Epoch 83: accuracy did not improve from 0.79367

Epoch 84: accuracy did not improve from 0.79367

Epoch 85: accuracy did not improve from 0.79367

Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

Epoch 88: accuracy did not improve from 0.79367

Epoch 89: accuracy did not improve from 0.79367

Epoch 90: accuracy did not improve from 0.79367

Epoch 91: accuracy did not improve from 0.79367

Epoch 92: accuracy did not improve from 0.79367

Epoch 93: accuracy did not improve from 0.79367

Epoch 94: accuracy did not improve from 0.79367

Epoch 95: accuracy did not improve from 0.79367

Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ..........batch_size=32, epochs=100, optimizer=adam; total time=  32.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367
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Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

Epoch 34: accuracy did not improve from 0.79367
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Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

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Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367

Epoch 51: accuracy did not improve from 0.79367

Epoch 52: accuracy did not improve from 0.79367

Epoch 53: accuracy did not improve from 0.79367

Epoch 54: accuracy did not improve from 0.79367

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Epoch 58: accuracy did not improve from 0.79367

Epoch 59: accuracy did not improve from 0.79367

Epoch 60: accuracy did not improve from 0.79367

Epoch 61: accuracy did not improve from 0.79367

Epoch 62: accuracy did not improve from 0.79367
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Epoch 63: accuracy did not improve from 0.79367

Epoch 64: accuracy did not improve from 0.79367

Epoch 65: accuracy did not improve from 0.79367

Epoch 66: accuracy did not improve from 0.79367

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Epoch 68: accuracy did not improve from 0.79367

Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

Epoch 71: accuracy did not improve from 0.79367

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Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

Epoch 88: accuracy did not improve from 0.79367

Epoch 89: accuracy did not improve from 0.79367

Epoch 90: accuracy did not improve from 0.79367
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Epoch 91: accuracy did not improve from 0.79367

Epoch 92: accuracy did not improve from 0.79367

Epoch 93: accuracy did not improve from 0.79367

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Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END .......batch_size=32, epochs=100, optimizer=rmsprop; total time=  31.2s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 17: accuracy did not improve from 0.79367
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Epoch 18: accuracy did not improve from 0.79367

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Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367
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Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

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Epoch 51: accuracy did not improve from 0.79367

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Epoch 69: accuracy did not improve from 0.79367

Epoch 70: accuracy did not improve from 0.79367

Epoch 71: accuracy did not improve from 0.79367

Epoch 72: accuracy did not improve from 0.79367

Epoch 73: accuracy did not improve from 0.79367
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Epoch 74: accuracy did not improve from 0.79367

Epoch 75: accuracy did not improve from 0.79367

Epoch 76: accuracy did not improve from 0.79367

Epoch 77: accuracy did not improve from 0.79367

Epoch 78: accuracy did not improve from 0.79367

Epoch 79: accuracy did not improve from 0.79367

Epoch 80: accuracy did not improve from 0.79367

Epoch 81: accuracy did not improve from 0.79367

Epoch 82: accuracy did not improve from 0.79367

Epoch 83: accuracy did not improve from 0.79367

Epoch 84: accuracy did not improve from 0.79367

Epoch 85: accuracy did not improve from 0.79367

Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

Epoch 88: accuracy did not improve from 0.79367

Epoch 89: accuracy did not improve from 0.79367

Epoch 90: accuracy did not improve from 0.79367

Epoch 91: accuracy did not improve from 0.79367

Epoch 92: accuracy did not improve from 0.79367

Epoch 93: accuracy did not improve from 0.79367

Epoch 94: accuracy did not improve from 0.79367

Epoch 95: accuracy did not improve from 0.79367

Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END .......batch_size=32, epochs=100, optimizer=rmsprop; total time=  31.3s

Epoch 1: accuracy did not improve from 0.79367
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Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

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Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

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Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367
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Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

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Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

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Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

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Epoch 50: accuracy did not improve from 0.79367

Epoch 51: accuracy did not improve from 0.79367

Epoch 52: accuracy did not improve from 0.79367

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Epoch 56: accuracy did not improve from 0.79367

Epoch 57: accuracy did not improve from 0.79367
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Epoch 58: accuracy did not improve from 0.79367

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Epoch 83: accuracy did not improve from 0.79367

Epoch 84: accuracy did not improve from 0.79367

Epoch 85: accuracy did not improve from 0.79367
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Epoch 86: accuracy did not improve from 0.79367

Epoch 87: accuracy did not improve from 0.79367

Epoch 88: accuracy did not improve from 0.79367

Epoch 89: accuracy did not improve from 0.79367

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Epoch 95: accuracy did not improve from 0.79367

Epoch 96: accuracy did not improve from 0.79367

Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END .......batch_size=32, epochs=100, optimizer=rmsprop; total time=  31.1s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367
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Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

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Epoch 18: accuracy did not improve from 0.79367

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Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=128, epochs=30, optimizer=adam; total time=   3.0s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367
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Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

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Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

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Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=128, epochs=30, optimizer=adam; total time=   3.0s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367
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Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

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Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=128, epochs=30, optimizer=adam; total time=   3.0s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367
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Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

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Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END .......batch_size=128, epochs=30, optimizer=rmsprop; total time=   2.8s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367
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Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END .......batch_size=128, epochs=30, optimizer=rmsprop; total time=   2.9s
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Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367
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Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END .......batch_size=128, epochs=30, optimizer=rmsprop; total time=   2.8s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367
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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

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Epoch 35: accuracy did not improve from 0.79367

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Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=128, epochs=50, optimizer=adam; total time=   5.1s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367
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Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367
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Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

Epoch 34: accuracy did not improve from 0.79367

Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=128, epochs=50, optimizer=adam; total time=   4.8s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367
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Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

Epoch 21: accuracy did not improve from 0.79367

Epoch 22: accuracy did not improve from 0.79367

Epoch 23: accuracy did not improve from 0.79367

Epoch 24: accuracy did not improve from 0.79367

Epoch 25: accuracy did not improve from 0.79367

Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

Epoch 34: accuracy did not improve from 0.79367

Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367
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Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367

Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=128, epochs=50, optimizer=adam; total time=   4.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367

Epoch 7: accuracy did not improve from 0.79367

Epoch 8: accuracy did not improve from 0.79367

Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367
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Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367

Epoch 18: accuracy did not improve from 0.79367

Epoch 19: accuracy did not improve from 0.79367

Epoch 20: accuracy did not improve from 0.79367

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Epoch 26: accuracy did not improve from 0.79367

Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367

Epoch 31: accuracy did not improve from 0.79367

Epoch 32: accuracy did not improve from 0.79367

Epoch 33: accuracy did not improve from 0.79367

Epoch 34: accuracy did not improve from 0.79367

Epoch 35: accuracy did not improve from 0.79367

Epoch 36: accuracy did not improve from 0.79367

Epoch 37: accuracy did not improve from 0.79367

Epoch 38: accuracy did not improve from 0.79367

Epoch 39: accuracy did not improve from 0.79367

Epoch 40: accuracy did not improve from 0.79367

Epoch 41: accuracy did not improve from 0.79367

Epoch 42: accuracy did not improve from 0.79367
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Epoch 43: accuracy did not improve from 0.79367

Epoch 44: accuracy did not improve from 0.79367

Epoch 45: accuracy did not improve from 0.79367

Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END .......batch_size=128, epochs=50, optimizer=rmsprop; total time=   4.5s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

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Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END .......batch_size=128, epochs=50, optimizer=rmsprop; total time=   4.5s

Epoch 1: accuracy did not improve from 0.79367

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Epoch 50: accuracy did not improve from 0.79367
[CV] END .......batch_size=128, epochs=50, optimizer=rmsprop; total time=   4.5s

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Epoch 100: accuracy did not improve from 0.79367
[CV] END .........batch_size=128, epochs=100, optimizer=adam; total time=   8.9s

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Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END .........batch_size=128, epochs=100, optimizer=adam; total time=   8.9s

Epoch 1: accuracy did not improve from 0.79367

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Epoch 100: accuracy did not improve from 0.79367
[CV] END .........batch_size=128, epochs=100, optimizer=adam; total time=   8.9s

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Epoch 100: accuracy did not improve from 0.79367
[CV] END ......batch_size=128, epochs=100, optimizer=rmsprop; total time=   8.5s

Epoch 1: accuracy did not improve from 0.79367

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Epoch 48: accuracy did not improve from 0.79367
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Epoch 49: accuracy did not improve from 0.79367

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Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ......batch_size=128, epochs=100, optimizer=rmsprop; total time=   8.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367
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Epoch 5: accuracy did not improve from 0.79367

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Epoch 61: accuracy did not improve from 0.79367

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Epoch 89: accuracy did not improve from 0.79367

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Epoch 97: accuracy did not improve from 0.79367

Epoch 98: accuracy did not improve from 0.79367

Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END ......batch_size=128, epochs=100, optimizer=rmsprop; total time=   8.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 15: accuracy did not improve from 0.79367
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Epoch 27: accuracy did not improve from 0.79367

Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=256, epochs=30, optimizer=adam; total time=   1.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

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Epoch 14: accuracy did not improve from 0.79367

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Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=256, epochs=30, optimizer=adam; total time=   1.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END ..........batch_size=256, epochs=30, optimizer=adam; total time=   1.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

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Epoch 9: accuracy did not improve from 0.79367

Epoch 10: accuracy did not improve from 0.79367

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Epoch 28: accuracy did not improve from 0.79367

Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END .......batch_size=256, epochs=30, optimizer=rmsprop; total time=   1.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

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Epoch 6: accuracy did not improve from 0.79367

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Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END .......batch_size=256, epochs=30, optimizer=rmsprop; total time=   1.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367
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Epoch 4: accuracy did not improve from 0.79367

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Epoch 29: accuracy did not improve from 0.79367

Epoch 30: accuracy did not improve from 0.79367
[CV] END .......batch_size=256, epochs=30, optimizer=rmsprop; total time=   1.7s
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Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

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Epoch 29: accuracy did not improve from 0.79367

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Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=256, epochs=50, optimizer=adam; total time=   2.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

Epoch 5: accuracy did not improve from 0.79367

Epoch 6: accuracy did not improve from 0.79367
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Epoch 7: accuracy did not improve from 0.79367

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Epoch 9: accuracy did not improve from 0.79367

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Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=256, epochs=50, optimizer=adam; total time=   2.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

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Epoch 11: accuracy did not improve from 0.79367
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Epoch 40: accuracy did not improve from 0.79367

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Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END ..........batch_size=256, epochs=50, optimizer=adam; total time=   2.9s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

Epoch 4: accuracy did not improve from 0.79367

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Epoch 10: accuracy did not improve from 0.79367

Epoch 11: accuracy did not improve from 0.79367

Epoch 12: accuracy did not improve from 0.79367

Epoch 13: accuracy did not improve from 0.79367

Epoch 14: accuracy did not improve from 0.79367

Epoch 15: accuracy did not improve from 0.79367

Epoch 16: accuracy did not improve from 0.79367

Epoch 17: accuracy did not improve from 0.79367
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Epoch 18: accuracy did not improve from 0.79367

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Epoch 46: accuracy did not improve from 0.79367

Epoch 47: accuracy did not improve from 0.79367

Epoch 48: accuracy did not improve from 0.79367

Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END .......batch_size=256, epochs=50, optimizer=rmsprop; total time=   2.7s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

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Epoch 22: accuracy did not improve from 0.79367
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Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END .......batch_size=256, epochs=50, optimizer=rmsprop; total time=   2.7s
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Epoch 1: accuracy did not improve from 0.79367

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Epoch 49: accuracy did not improve from 0.79367

Epoch 50: accuracy did not improve from 0.79367
[CV] END .......batch_size=256, epochs=50, optimizer=rmsprop; total time=   2.6s

Epoch 1: accuracy did not improve from 0.79367

Epoch 2: accuracy did not improve from 0.79367

Epoch 3: accuracy did not improve from 0.79367

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Epoch 100: accuracy did not improve from 0.79367
[CV] END .........batch_size=256, epochs=100, optimizer=adam; total time=   5.0s

Epoch 1: accuracy did not improve from 0.79367

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Epoch 99: accuracy did not improve from 0.79367

Epoch 100: accuracy did not improve from 0.79367
[CV] END .........batch_size=256, epochs=100, optimizer=adam; total time=   5.0s
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Epoch 1: accuracy did not improve from 0.79367

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Epoch 100: accuracy did not improve from 0.79367
[CV] END .........batch_size=256, epochs=100, optimizer=adam; total time=   5.0s

Epoch 1: accuracy did not improve from 0.79367

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[CV] END ......batch_size=256, epochs=100, optimizer=rmsprop; total time=   4.8s

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[CV] END ......batch_size=256, epochs=100, optimizer=rmsprop; total time=   4.8s

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[CV] END ......batch_size=256, epochs=100, optimizer=rmsprop; total time=   4.8s

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In [9]:

```python
# 7. Print results
# ----------------------------
print(f"\n Best Accuracy: {grid_result.best_score_:.4f}")
print(f" Best Parameters: {grid_result.best_params_}")

print("\n All Grid Search Results:")
means = grid_result.cv_results_['mean_test_score']
stds = grid_result.cv_results_['std_test_score']
params = grid_result.cv_results_['params']

for mean, std, param in zip(means, stds, params):
    print(f"{mean:.4f} (+/- {std:.4f}) with: {param}")
```

```
Best Accuracy: 0.7484
Best Parameters: {'batch_size': 256, 'epochs': 30, 'optimizer': 'adam'}

All Grid Search Results:
0.7369 (+/- 0.0026) with: {'batch_size': 16, 'epochs': 30, 'optimizer': 'adam'}
0.7435 (+/- 0.0031) with: {'batch_size': 16, 'epochs': 30, 'optimizer': 'rmsprop'}
0.7345 (+/- 0.0066) with: {'batch_size': 16, 'epochs': 50, 'optimizer': 'adam'}
0.7447 (+/- 0.0055) with: {'batch_size': 16, 'epochs': 50, 'optimizer': 'rmsprop'}
0.7273 (+/- 0.0022) with: {'batch_size': 16, 'epochs': 100, 'optimizer': 'adam'}
0.7364 (+/- 0.0040) with: {'batch_size': 16, 'epochs': 100, 'optimizer': 'rmsprop'}
0.7400 (+/- 0.0070) with: {'batch_size': 32, 'epochs': 30, 'optimizer': 'adam'}
0.7411 (+/- 0.0036) with: {'batch_size': 32, 'epochs': 30, 'optimizer': 'rmsprop'}
0.7365 (+/- 0.0068) with: {'batch_size': 32, 'epochs': 50, 'optimizer': 'adam'}
0.7370 (+/- 0.0038) with: {'batch_size': 32, 'epochs': 50, 'optimizer': 'rmsprop'}
0.7332 (+/- 0.0027) with: {'batch_size': 32, 'epochs': 100, 'optimizer': 'adam'}
0.7326 (+/- 0.0041) with: {'batch_size': 32, 'epochs': 100, 'optimizer': 'rmsprop'}
0.7429 (+/- 0.0025) with: {'batch_size': 128, 'epochs': 30, 'optimizer': 'adam'}
0.7471 (+/- 0.0019) with: {'batch_size': 128, 'epochs': 30, 'optimizer': 'rmsprop'}
0.7406 (+/- 0.0047) with: {'batch_size': 128, 'epochs': 50, 'optimizer': 'adam'}
0.7410 (+/- 0.0056) with: {'batch_size': 128, 'epochs': 50, 'optimizer': 'rmsprop'}
0.7343 (+/- 0.0043) with: {'batch_size': 128, 'epochs': 100, 'optimizer': 'adam'}
0.7353 (+/- 0.0011) with: {'batch_size': 128, 'epochs': 100, 'optimizer': 'rmsprop'}
0.7484 (+/- 0.0034) with: {'batch_size': 256, 'epochs': 30, 'optimizer': 'adam'}
0.7466 (+/- 0.0052) with: {'batch_size': 256, 'epochs': 30, 'optimizer': 'rmsprop'}
0.7453 (+/- 0.0045) with: {'batch_size': 256, 'epochs': 50, 'optimizer': 'adam'}
0.7469 (+/- 0.0040) with: {'batch_size': 256, 'epochs': 50, 'optimizer': 'rmsprop'}
0.7426 (+/- 0.0029) with: {'batch_size': 256, 'epochs': 100, 'optimizer': 'adam'}
0.7384 (+/- 0.0051) with: {'batch_size': 256, 'epochs': 100, 'optimizer': 'rmsprop'}
```

6.2 Grid search Results summary

Batch Size	Epochs	Optimizer	Accuracy (Mean ± Std)
256	30	Adam	**0.7484 ± 0.0034** *(Best)*
128	30	RMSprop	0.7471 ± 0.0019
256	50	RMSprop	0.7469 ± 0.0040
256	30	RMSprop	0.7466 ± 0.0052
128	30	Adam	0.7429 ± 0.0025
16	50	RMSprop	0.7447 ± 0.0055
16	30	RMSprop	0.7435 ± 0.0031
32	30	RMSprop	0.7411 ± 0.0036

The best performing configuration was: Batch Size = 256, Epochs = 30, Optimizer = Adam
Achieved Accuracy: 0.7484

6.3 Performance Insights

1. Batch size: Larger batch sizes(128 and 256) consistently outperformed smaller batch

sizes. mainly batch size 256 which yielded the best performance, likely due to more stable gradient updates and efficient hardware utilization.

2. Epochs: Peak accuracy was usually achieved within 30 to 50 epochs beyond which performance was likely plateaued or slightly dropped. This confirms that early stopping was effective in reducing overtraining risk.
3. Optimizer comparison: Adam had slight edge in stability and generalization. Whereas RMSprop performed closely especially with smaller batch sizes but was marginally less consistent across epochs
4. Weight Initialization: Using henormal contributed to faster convergence and better handling of ReLU activations therefore supporting the deeper architecture's stability.

After performing Grid search I observed a clear trend : Adam with batch size 256 and early stopping at roughly 30 epochs consistently gives the highest validation accuracy.

The model benefits from HeNormal initializer and both Adam and RMSprop are effective optimizers depending on batch size.

This hyperparameter tuning phase played a critical role in identifying a configuration that balances accuracy, stability and training efficiency. The selected best model will be used in further evaluation and deployment stages.